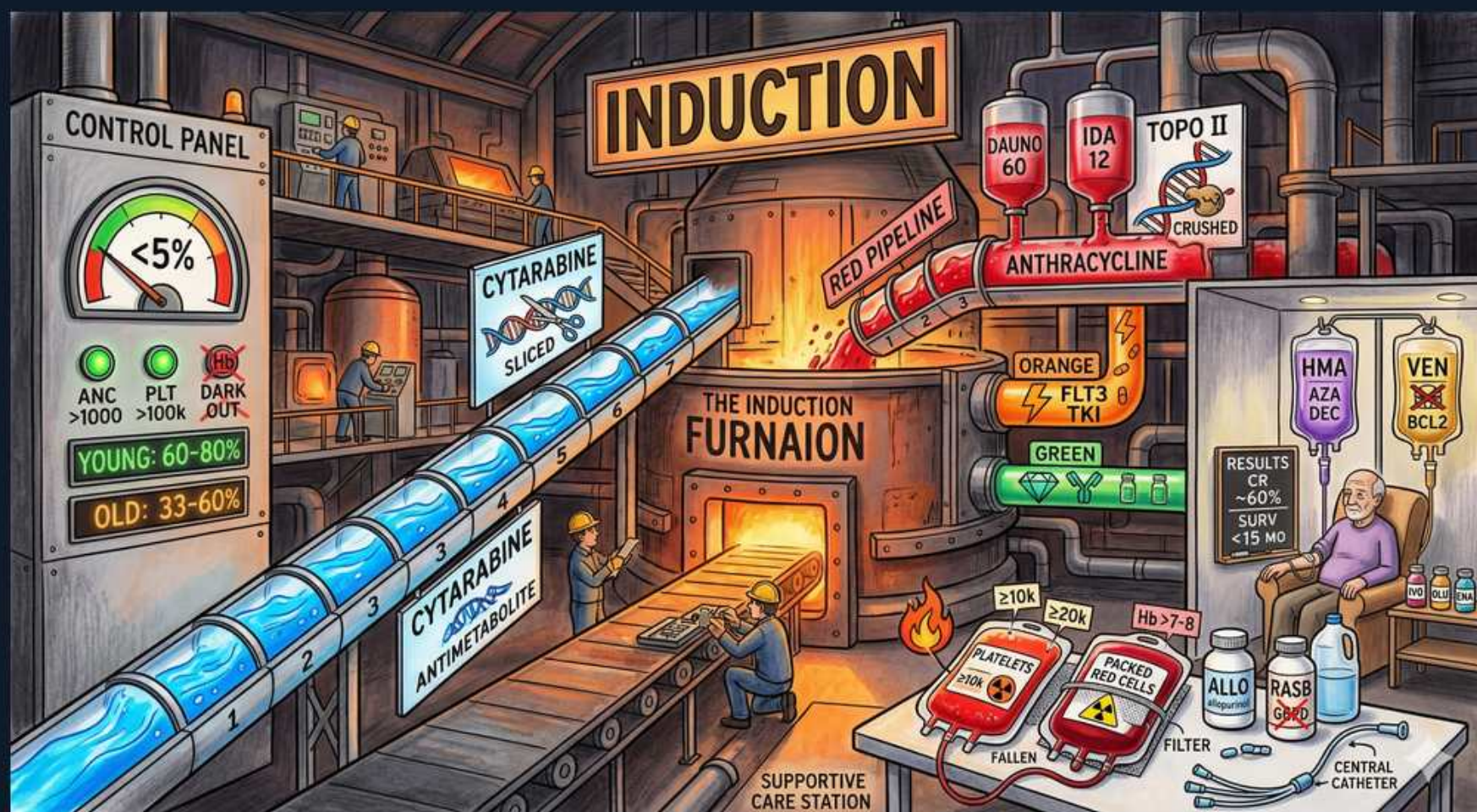


## AML — Induction (7+3), Targeted Add-ons & Supportive Care





1. INDUCTION furnace

WHAT YOU SEE

'INDUCTION' sign over a glowing furnace, 'THE INDUCTION FURNACE'.

WHAT IT ENCODES

Standard intensive induction is '7+3': 7 days continuous-infusion cytarabine plus 3 days of an anthracycline. Goal is morphologic complete remission (<5% marrow blasts with count recovery).

BOARD TRAP

Induction aims for CR, not cure — every patient needs post-remission consolidation. A marrow at day 14 may still show blasts and prompt reinduction.

2. Cytarabine pipeline

WHAT YOU SEE

Blue 'CYTARABINE' pipeline labeled 'ANTIMETABOLITE', '7 DAYS'.

WHAT IT ENCODES

Cytarabine (ara-C) is a pyrimidine antimetabolite given as a 7-day continuous infusion — the backbone of induction.

BOARD TRAP

Cytarabine is the antimetabolite (S-phase); the anthracycline is the topoisomerase II / intercalator. Don't swap their mechanisms.

3. Anthracycline 3 days

WHAT YOU SEE

Red 'ANTHRACYCLINE' pipeline, 'TOPO II', 'CRUSHED' DNA.

WHAT IT ENCODES

Daunorubicin or idarubicin for 3 days. Anthracyclines inhibit topoisomerase II and intercalate DNA, generating strand breaks.

BOARD TRAP

Anthracycline = cardiotoxicity (cumulative, dose-dependent dilated cardiomyopathy); check baseline ejection fraction. This is the topo II agent, not the antimetabolite.

4. Dauno 60 / Ida 12

WHAT YOU SEE

'DAUNO 60' and 'IDA 12' dose tiles.

WHAT IT ENCODES

Typical induction anthracycline dosing: daunorubicin 60–90 mg/m2 or idarubicin 12 mg/m2 for 3 days. Higher daunorubicin doses improved outcomes in younger adults.

BOARD TRAP

These are induction anthracycline doses, not cytarabine doses. Standard induction ara-C (100–200 mg/m2/day) differs from high-dose ara-C used in consolidation.

5. <5% blasts gauge

WHAT YOU SEE

Control panel reading '<5%'.

WHAT IT ENCODES

Complete remission requires <5% marrow blasts plus neutrophil and platelet recovery and no extramedullary disease. The CR target after induction.

BOARD TRAP

<5% blasts defines remission; ≥20% blasts defines the original AML diagnosis. Don't mix the diagnostic threshold with the remission threshold.

6. CR rates young/old

WHAT YOU SEE

'YOUNG 60-80%' / 'OLD 33-60%' CR bands.

WHAT IT ENCODES

Complete-remission rates are higher in younger fit adults than in older patients, who have more adverse genetics and worse tolerance of intensive therapy.

BOARD TRAP

Older / unfit patients are often better served by venetoclax + azacitidine than by intensive 7+3 — fitness, not age alone, drives the choice.

7. FLT3 TKI add-on

WHAT YOU SEE

Orange 'FLT3 B / TKI' tile.

WHAT IT ENCODES

For FLT3-ITD or FLT3-TKD mutated AML, add midostaurin to 7+3 induction and consolidation — improves overall survival.

BOARD TRAP

Midostaurin is the frontline FLT3 add-on; gilteritinib is for relapsed/refractory FLT3 disease. Match drug to setting.

8. HMA / AZA / DEC

WHAT YOU SEE

'HMA — AZA / DEC' panel.

WHAT IT ENCODES

Hypomethylating agents (azacitidine, decitabine), usually with venetoclax, are the standard for older/unfit patients and a backbone in TP53 disease.

BOARD TRAP

HMA + venetoclax is the non-intensive standard; it is not equivalent to 7+3 and is chosen for unfit patients, not as a 'milder induction' for fit ones.

9. VEN / BCL2

WHAT YOU SEE

'VEN — BCL2' panel, 'RESULTS CR ~60%'.

WHAT IT ENCODES

Venetoclax is a BCL-2 inhibitor; with azacitidine it yields high CR rates in older patients. Requires tumor lysis prophylaxis and a ramp-up dose schedule.

BOARD TRAP

Venetoclax in AML still carries tumor lysis risk — use ramp-up dosing and TLS prophylaxis, especially with high blast burden.

## 10. Supportive — transfusion

### WHAT YOU SEE

'SUPPORTIVE CARE STATION' with blood bags, 'Hb>7-8', '<10K / <20K'.

### WHAT IT ENCODES

Supportive care: transfuse RBCs for hemoglobin ~7–8, platelets prophylactically below 10k (or below 20k with fever/bleeding risk). Aggressive hydration and TLS management.

### BOARD TRAP

Prophylactic platelet threshold in stable AML is 10k; the higher 20k/50k thresholds apply with fever, bleeding, or procedures — don't over-transfuse.

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## 11. Neutropenic fever / ALLO

### WHAT YOU SEE

'ALLO' and antibiotic/'RASB' items near a central catheter.

### WHAT IT ENCODES

Neutropenic fever is a medical emergency: start broad-spectrum anti-pseudomonal antibiotics (e.g. cefepime/piperacillin-tazobactam) empirically. Rasburicase for high-risk tumor lysis with elevated uric acid.

### BOARD TRAP

In febrile neutropenia, begin empiric antibiotics immediately — do not wait for culture results. Rasburicase is contraindicated in G6PD deficiency.

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## 12. Central catheter

### WHAT YOU SEE

'CENTRAL CATHETER' on the right.

### WHAT IT ENCODES

A central venous catheter is placed for continuous cytarabine infusion, transfusions, and antibiotics during the prolonged cytopenic period.

### BOARD TRAP

The line itself is an infection source — line-associated bacteremia must be considered in any neutropenic fever workup.

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